

Notebook 01: ODE Model Simulation

Implements the deterministic 5-equation mean-field system, sweeps parameter space, and maps the stability boundary.

Important Project Safety Notice

Before executing or citing the findings in this notebook, please read the public guidance on what this project is and is not claiming:

[docs/not_saying.md](#) - *What This Theory Is NOT Claiming*

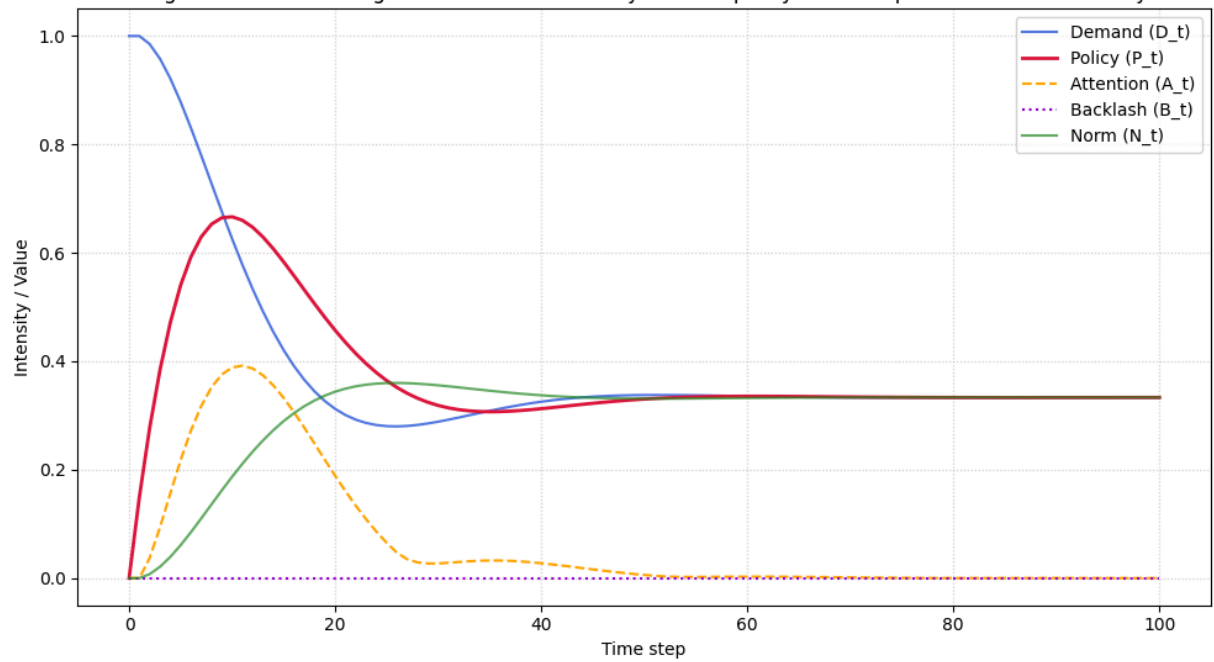
```
In [1]: import sys
sys.path.append('.')
from src.pendulum_model import PendulumSimulation
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

1. Run Standard Scenarios

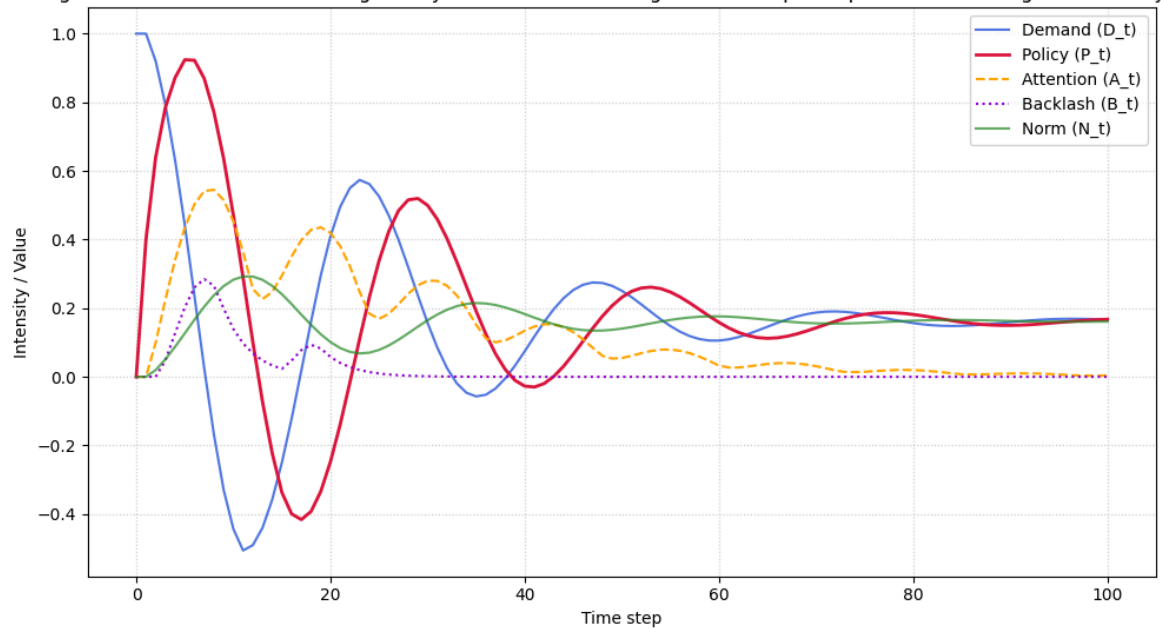
We load pre-configured scenarios representing different theoretical regimes.

```
In [2]: scenarios = ['stable_convergence', 'sustained_oscillation', 'punctuated_sawtooth']
for name in scenarios:
    sim = PendulumSimulation.from_scenario(name)
    sim.run(steps=100)
    fig = sim.plot_time_series(title=f'Scenario: {name}')
    plt.show()
```

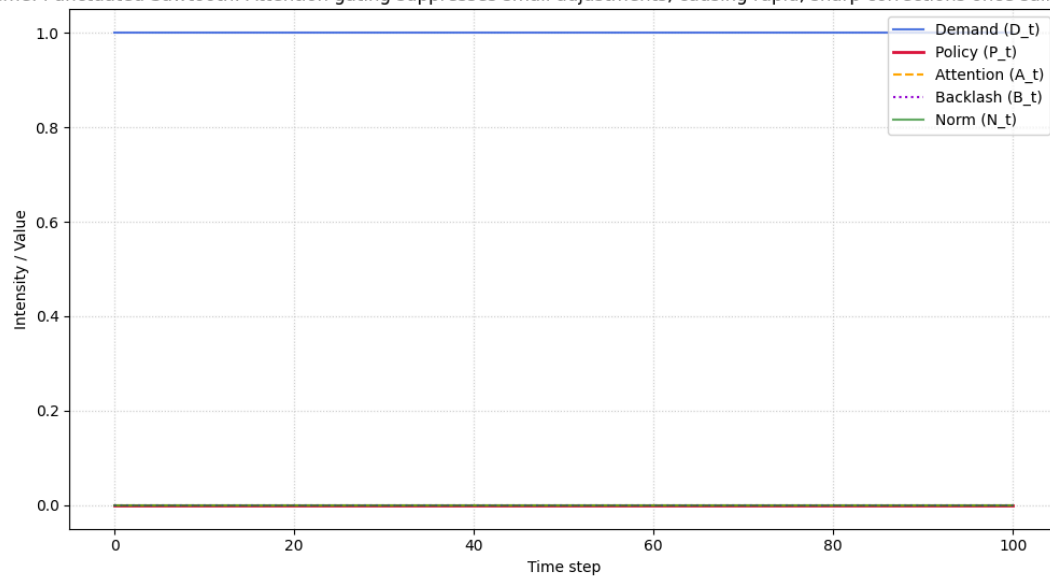
Scenario: stable_convergence
Regime: Stable Convergence: No feedback delay ensures policy matches public demand smoothly.



Scenario: sustained_oscillation
Regime: Sustained Oscillation: High delay combined with strong correction speeds produces recurring overshoot cycles.



Scenario: punctuated_sawtooth
 Regime: Punctuated Sawtooth: Attention gating suppresses small adjustments, causing rapid, sharp corrections once salience is high.



2. Stability Boundary Parameter Sweep

We sweep over τ (delay lag) and λ (correction speed) to map the boundary between stable convergence and cycles.

```
In [3]: sim = PendulumSimulation(alpha=0.2, nu=0.05)
df_sweep = sim.parameter_sweep(tau_range=list(range(6)), lambda_range=list(np.linspace(0.05, 0.9, 100)))
print(df_sweep.head(10))
```

	tau	lambda	amplitude_std	sign_changes	final_value	regime
0	0	0.050000	5.263846e-03	3	0.192636	convergent
1	0	0.144444	9.647274e-04	5	0.189568	convergent
2	0	0.238889	8.186878e-05	6	0.188798	convergent
3	0	0.333333	3.216363e-06	8	0.188651	convergent
4	0	0.427778	2.442961e-07	8	0.188743	convergent
5	0	0.522222	4.523163e-09	6	0.188937	convergent
6	0	0.616667	7.179119e-11	5	0.189174	convergent
7	0	0.711111	1.833649e-12	2	0.189423	convergent
8	0	0.805556	8.589483e-15	1	0.189669	convergent
9	0	0.900000	1.980222e-16	0	0.189903	convergent

```
In [4]: # 1. Run numerical sweep pivot
pivot = df_sweep.pivot(index='lambda', columns='tau', values='amplitude_std')

# 2. Run analytical stability sweep over a finer grid for a smooth boundary
df_analytical = sim.analytical_stability_sweep(
    tau_range=list(range(6)),
    lambda_range=list(np.linspace(0.05, 0.9, 100))
)
pivot_analytical = df_analytical.pivot(index='lambda', columns='tau', values='spect

# 3. Plot stability boundary map
plt.figure(figsize=(8, 6))
# Plot numerical sweep as heatmap
plt.imshow(pivot, aspect='auto', origin='lower', extent=[0, 5, 0.05, 0.9], cmap='ma
```

```
plt.colorbar(label='Policy standard deviation (amplitude)')

# Overlay analytical stability boundary (contour where spectral radius = 1.0)
tau_grid = pivot_analytical.columns
lambda_grid = pivot_analytical.index
plt.contour(
    tau_grid, lambda_grid, pivot_analytical.values,
    levels=[1.0], colors='white', linewidths=2.5, linestyle='--',
    label='Analytical Boundary (spectral radius = 1.0)'
)

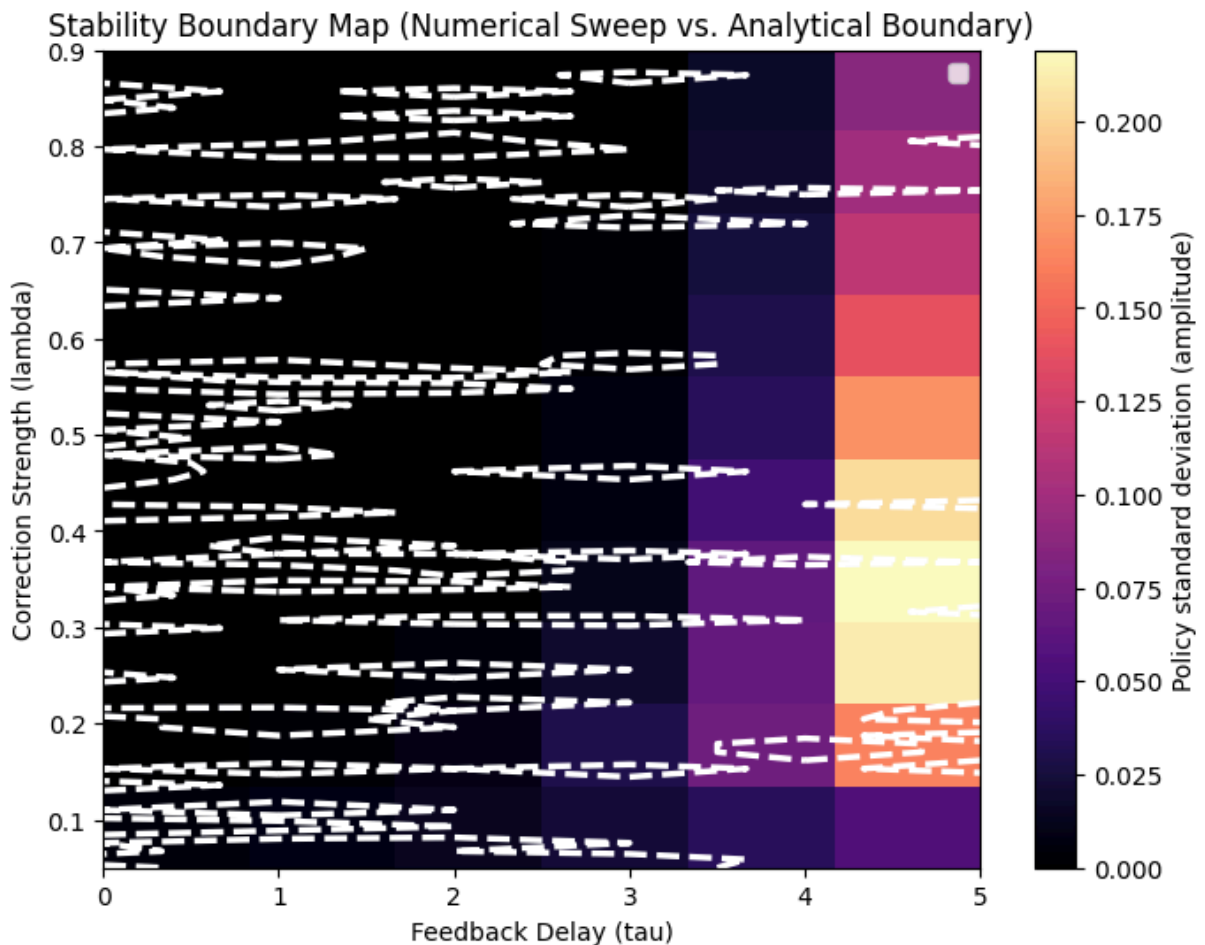
plt.xlabel('Feedback Delay (tau)')
plt.ylabel('Correction Strength (lambda)')
plt.title('Stability Boundary Map (Numerical Sweep vs. Analytical Boundary)')
plt.legend(loc='upper right')
plt.show()
```

C:\Users\admir\AppData\Local\Temp\ipykernel_2300\638988088.py:20: UserWarning: The following kwargs were not used by contour: 'label'

```
plt.contour(
```

C:\Users\admir\AppData\Local\Temp\ipykernel_2300\638988088.py:29: UserWarning: No artists with labels found to put in legend. Note that artists whose label start with an underscore are ignored when legend() is called with no argument.

```
plt.legend(loc='upper right')
```



In []:

3. The Theory-to-Empirics Bridge

It is important to clarify how this mathematical Ordinary Differential Equation (ODE) model connects to the empirical econometric analysis in the subsequent notebooks:

1. **Generative Theoretical Model:** The 5-equation ODE model is used as a generative theoretical framework to explore the system's qualitative dynamics (such as stable convergence, sustained oscillations, and punctuated sawtooth patterns). We do **not** directly estimate or calibrate the structural ODE parameters from the state-year panel data.
2. **Reduced-Form Implications:** The empirical regression models in Notebook 07 and 08 test the reduced-form implications of this theoretical feedback structure:
 - **H1 (Thermostatic Feedback):** Tests whether policy intensity level predicts later backlash (policy-to-backlash link) and whether backlash predicts a policy correction (backlash-to-policy rollback link).
 - **H7 (Institutional Dampening):** Tests if federal waivers act as safety valves that dampen these dynamic links.
3. **Parameter Mapping:** While the ODE uses continuous unbounded variables ($P_t, N_t \in [-2.0, 2.0]$), the empirical panel indicators are standardized composites. The theoretical simulation acts as a conceptual roadmap rather than a direct statistical fit.